

Bitcoin Volatility Prediction

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CAPSTONE PROJECT

AGENDA

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INTRODUCTION

Bitcoin swings hard, in bursts. This project predicts how big the swings will be — not which way price moves.



Nobody knows how rough next
week will be.

WHAT WE BUILT



A Turbulence Forecast

Predicts how much Bitcoin will move over the next week — not which direction.



Tested Like Real Trading

Checked against months of brand-new data the model never saw while learning.



Ready to Use

Built to plug straight into real risk and pricing decisions.

OUR DATA



8 Years of History

Hourly Bitcoin trading data going back to 2017 — over 70,000 hours in total.

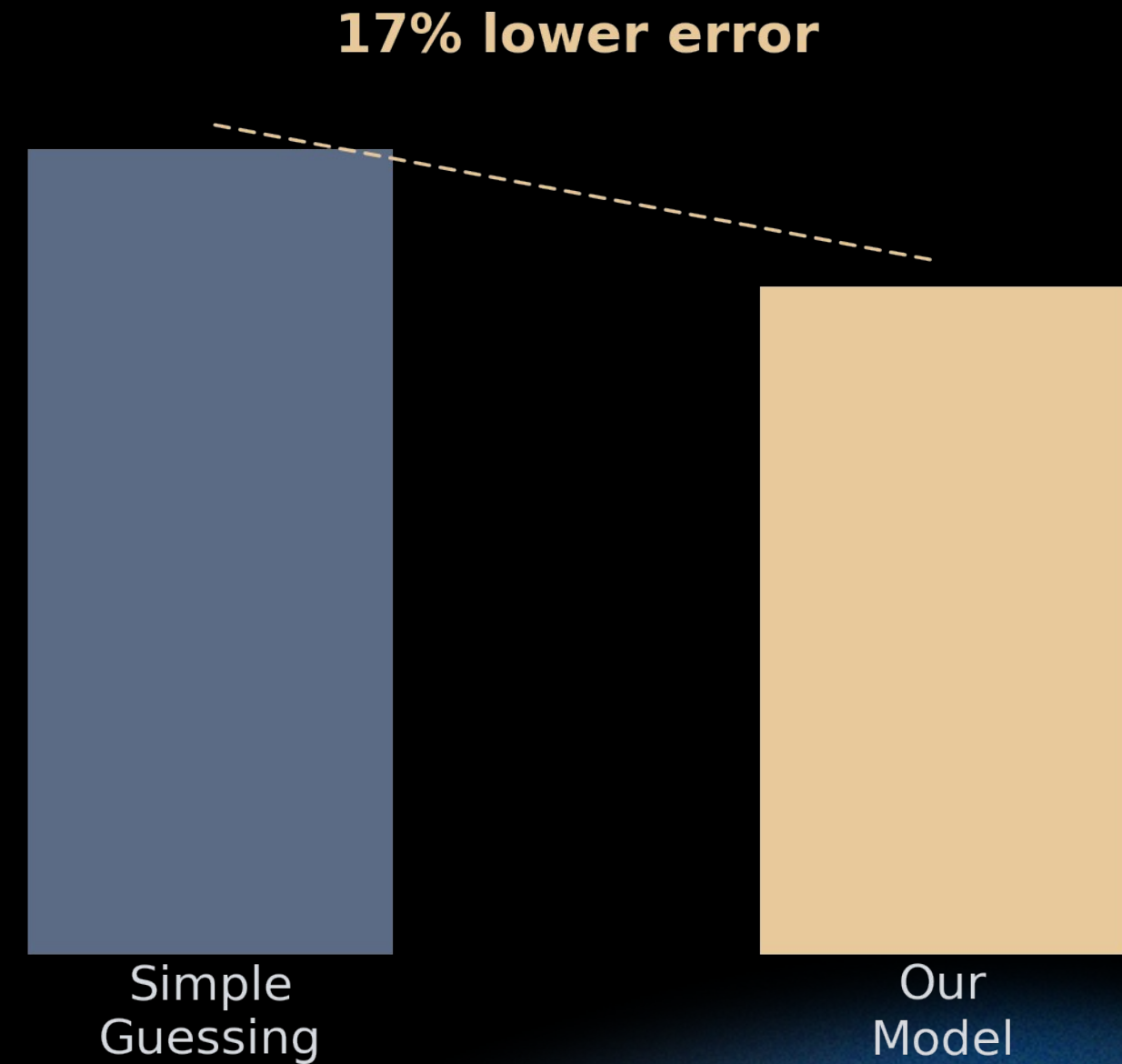
Calm, Then Chaos

Long quiet stretches, then sudden bursts. That pattern is exactly what the model learns to predict.

THE RESULTS

Our best model beat simple guessing by 17%.

And it kept working on brand-new data it had never seen before — the real test of whether a model actually works.



OUR MODELS

Classic

GARCH

The textbook approach to volatility. Struggled at this longer forecast horizon.

Winner

HAR-RV-X

Combines short-, medium-, and long-term trends. Our best performer, on old and brand-new data alike.

Runner-Up

EWMA

A simpler, industry-standard method. Solid, but not as sharp as our winner.



LIVE TESTING

The real test: brand-new Bitcoin data the model had never seen, scored the way a trader actually would — before the outcome was known.

COLLECT

PREDICT

COMPARE

REAL-WORLD USE

01

Sizing a Trade

If the model expects a bigger swing, a trader risks less money on that trade — and gets a clear estimate of how much they could lose in a bad week.

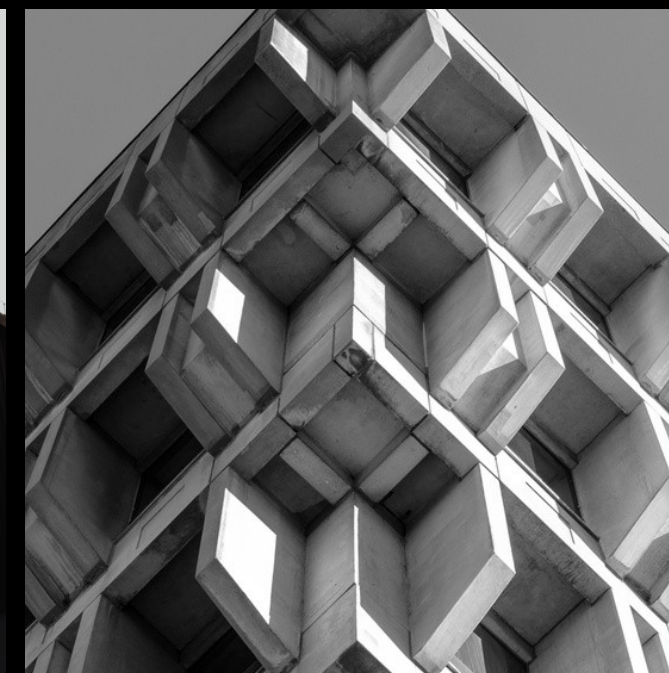
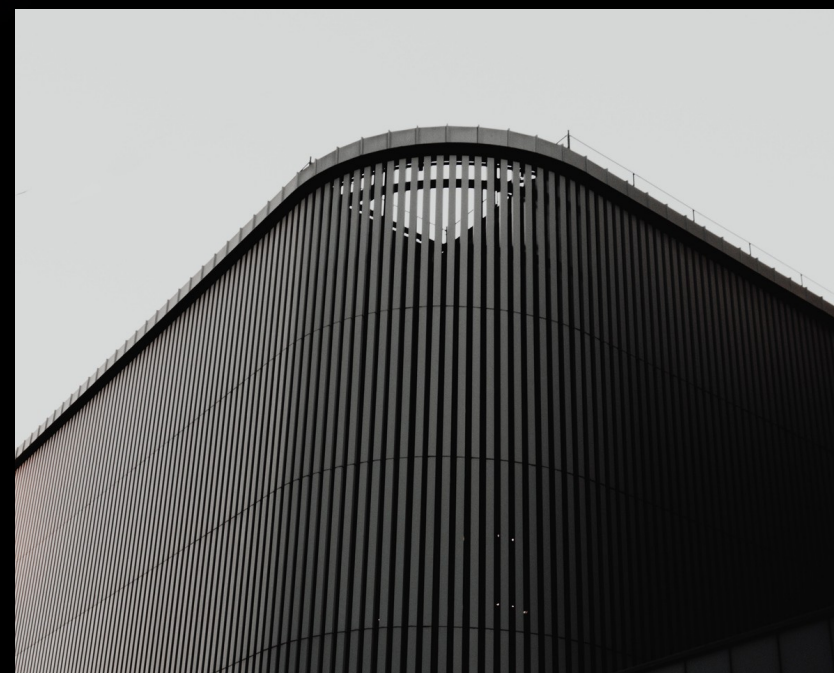


REAL-WORLD USE

02

Pricing an Option

Option prices depend on how much a market is expected to move. Our forecast gives a data-driven answer to compare against what the market is charging.





MARKET NEWS

Price data alone can't see
breaking news or new regulation
— a real limit on any forecast like
this.



ONE COIN ONLY

This model only looks at Bitcoin. It
doesn't yet use signals from other
cryptocurrencies.

GOOD TO KNOW

Two honest caveats

No model is perfect. Two things are worth knowing before
using this one.

 QUESTIONS?

THANK YOU

Cohort
DSB2

Project
Bitcoin Volatility Prediction

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